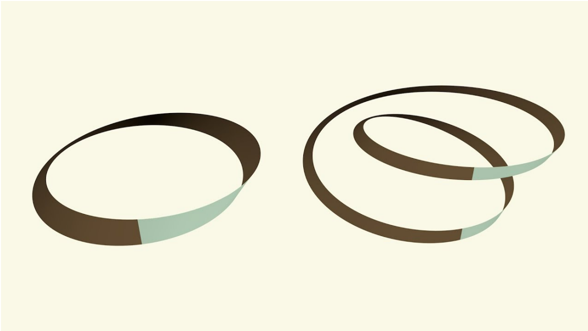
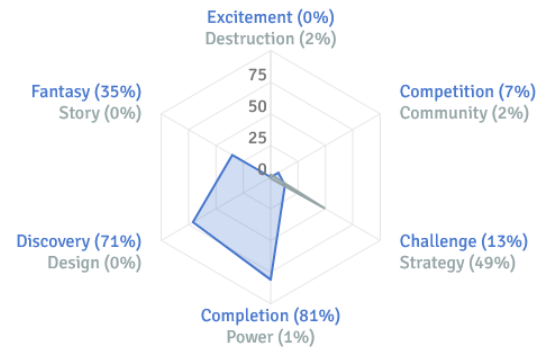
Möbius Magnus - Overview

Main Inspiration:

Möbius Magnus is a game that experiments with Möbius strips and other two dimensional manifolds in a three dimensional environment. Many of them are non-orientable shapes, meaning they have twist that entangles their top & bottom side.



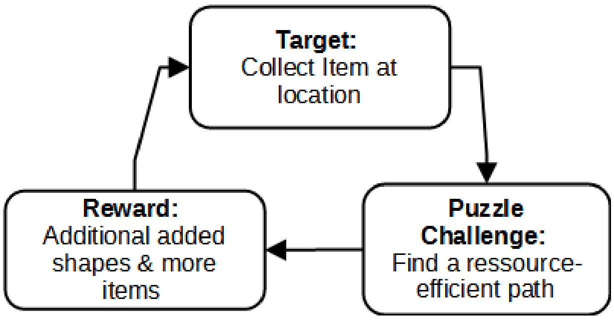


General Information

GENRE: 3D Puzzle
AGE RATING: Everyone (E)
PLATFORM: PC
Gaming motivation of the target audience to the left

Game Concept / Mechanics:

In “Möbius Magnus” the player tries to create the biggest possible entanglement of shapes. To do so they have to manage movement resources with which they collect items, that either give new resources or spawn more shapes.



Visual Concept

The game’s setting experiments with bending architecture in a setting in space; However unlike space it is not completely dark, but rather calming with pale colors.

Team:

Game Design: Valentin Then Bergh (Project Lead),
Anthony Huaman (Left before the project finished)
Game Art: Sara Alkotkat (Character), Aleksandr Maldon (Environment)
Game Programming: Alla-Edine Rakik, Ilia Kutsnezov
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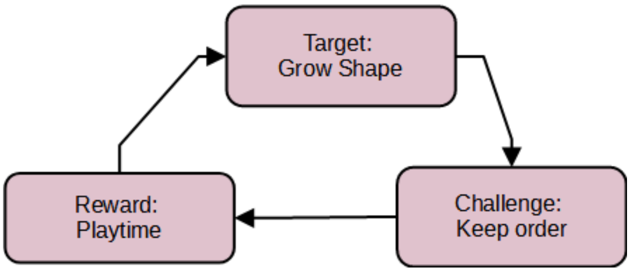
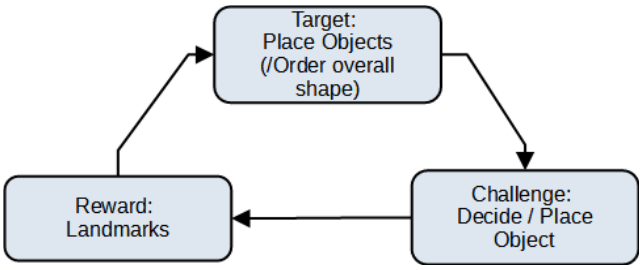
Game Loops

Simplified game-loops

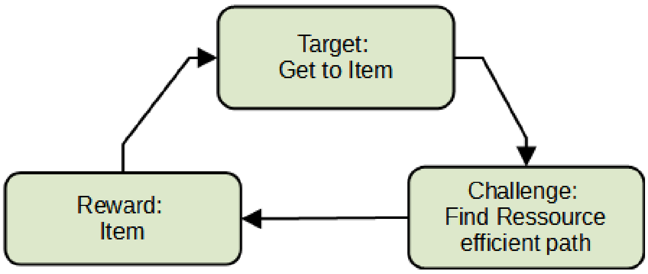
Overarching Loop = what the game is overall about



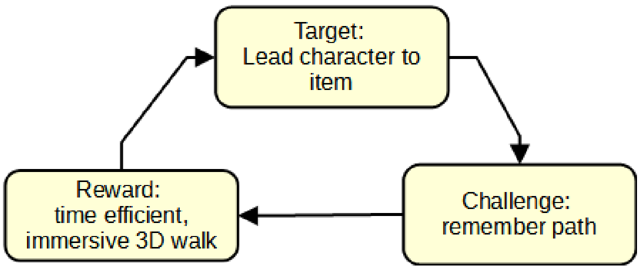
Tactical Loop = How the player makes & adjusts their tactic (=how they manage the overarching challenge)



Action Loop = What the player actively does / thinks throughout most of the game

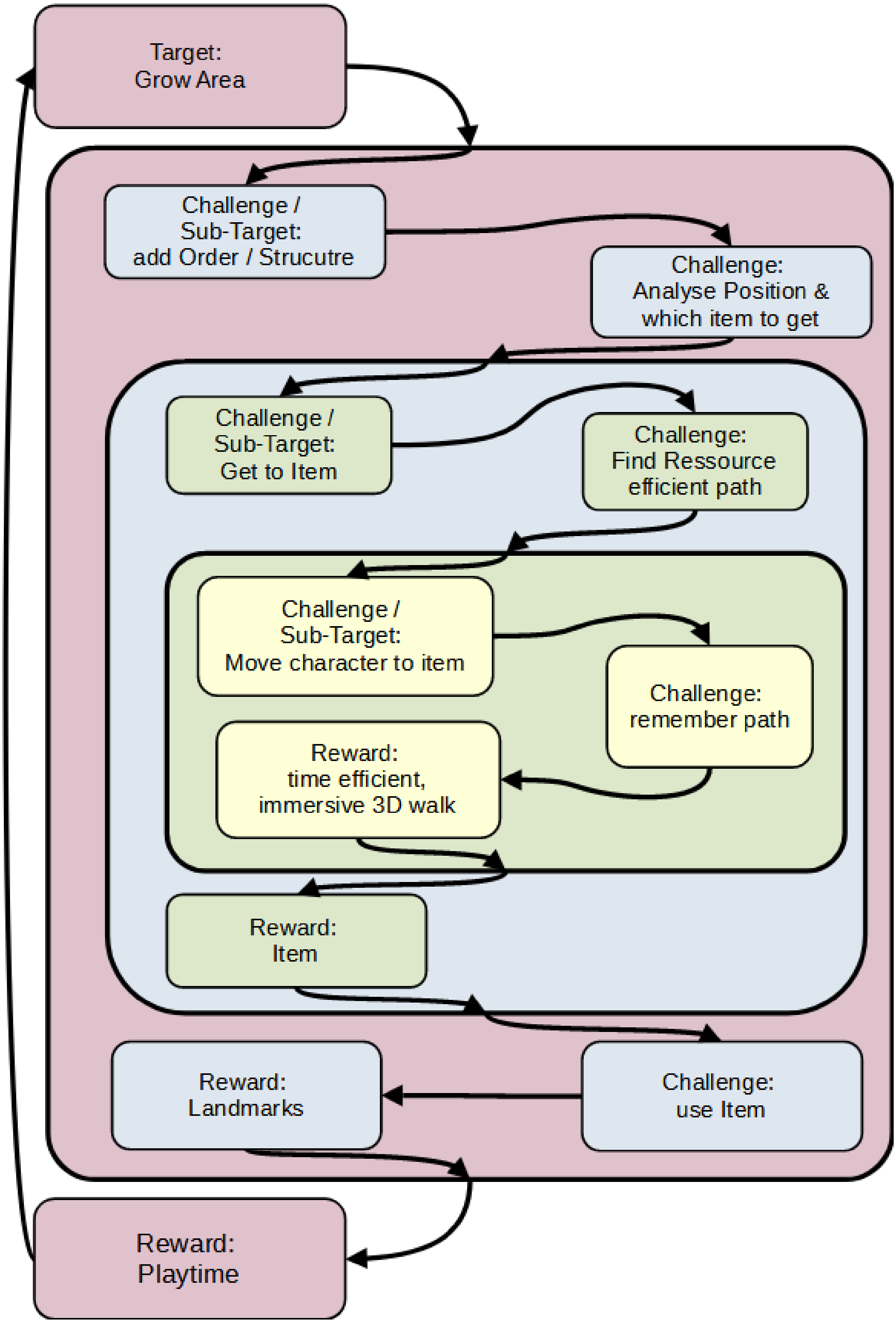


Atomic Loop = What happens on a moment by moment basis (=what the player subconsciously does)



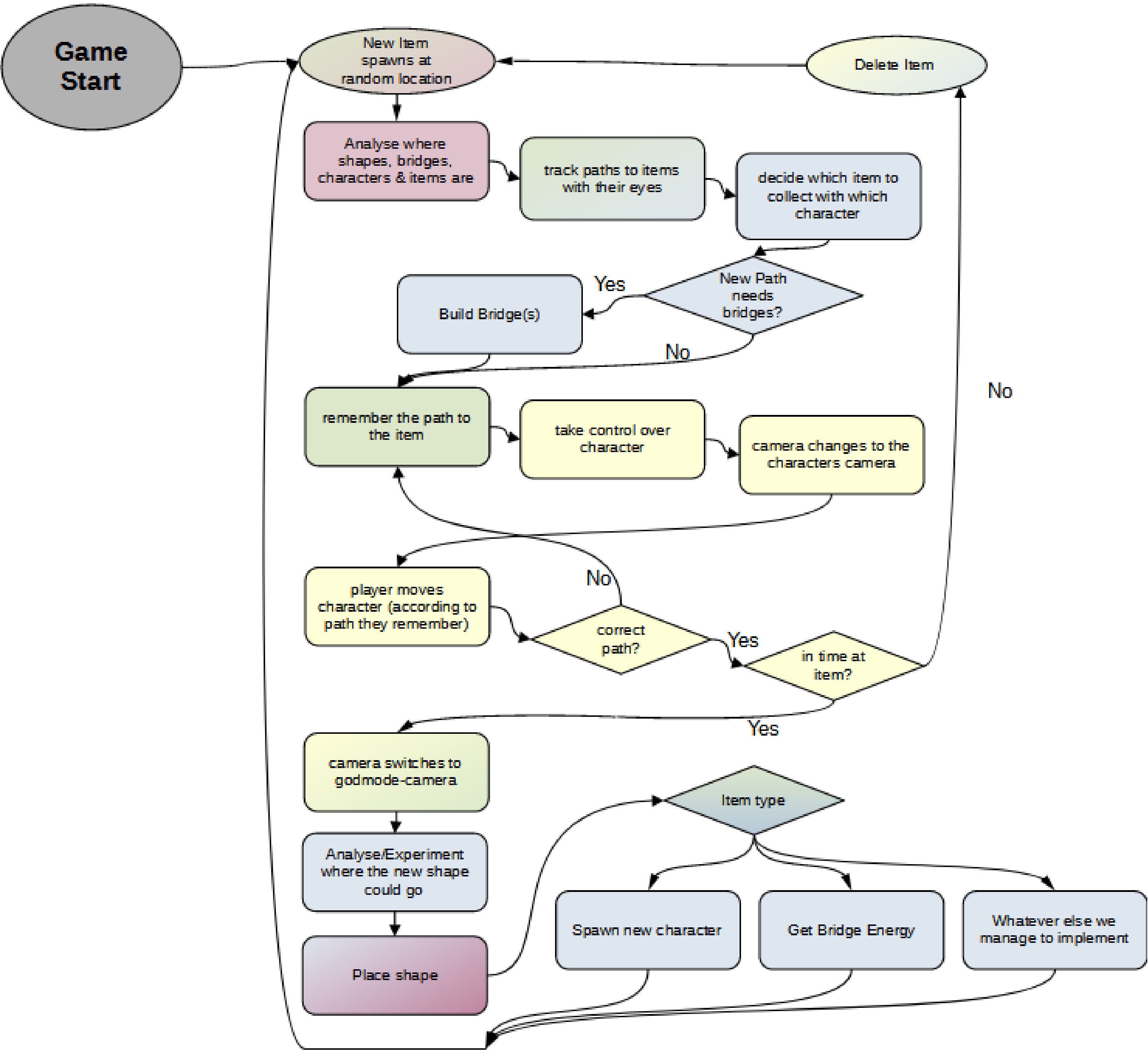
All of the simplified loops connect together into the

Full game loop



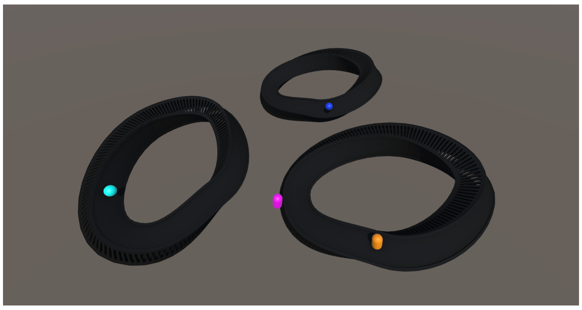
Player Perspective Loop

As players don't think in terms of game-loop, I've created a player-action flow diagram (the colors correspond to the game-loop colors)

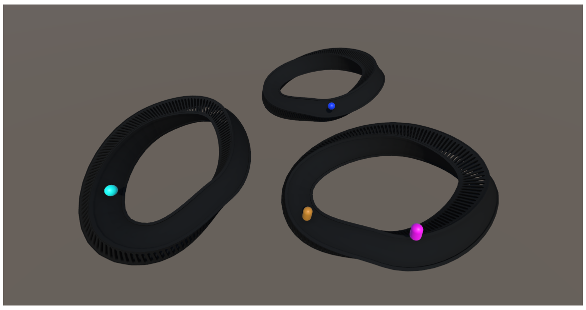


Circles = computer does this
Diamond = conditional progression through the loop

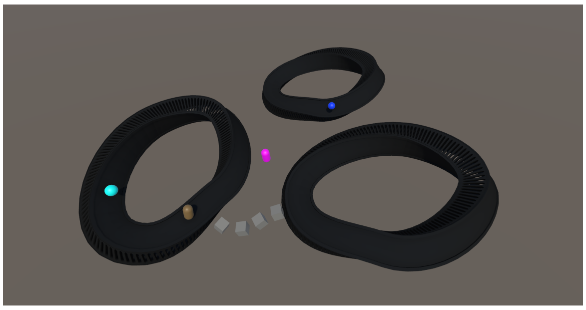
Gameplay Example:



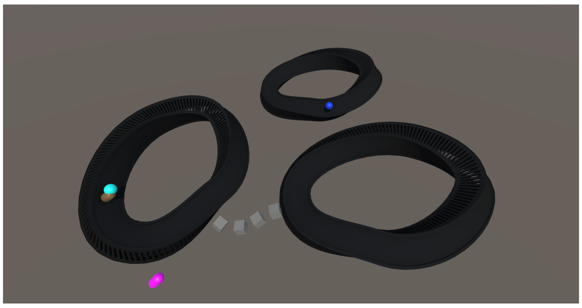
The game start:
Each shape checks whether it should spawn an energy- or new-shape-item



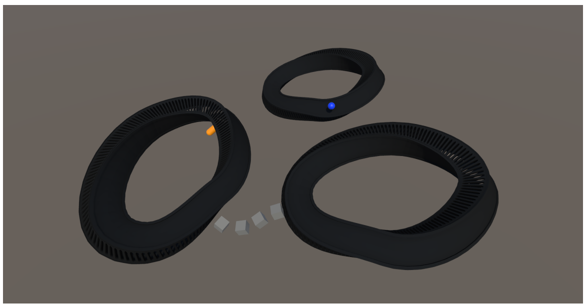
the player takes control over the character (loss of energy) turns it around (small loss of energy)



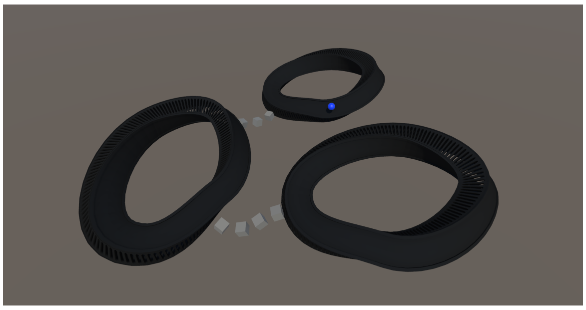
The player builds a bridge (big loss of energy)



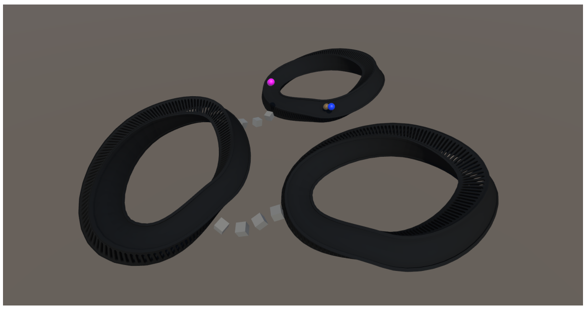
The character collects the energy item (+energy)



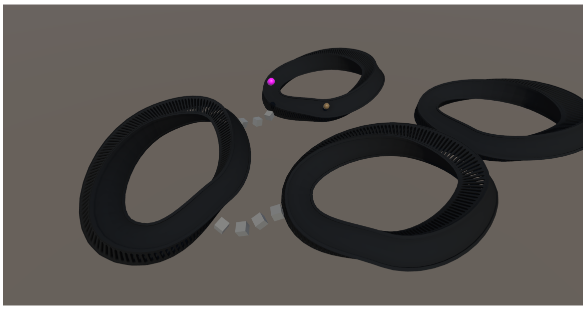
the player speed towards the next item



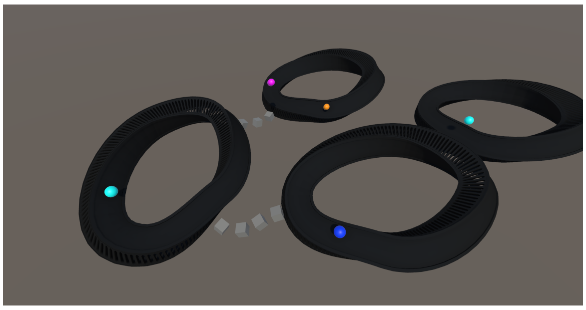
builds a new bridge (characters energy reduced)



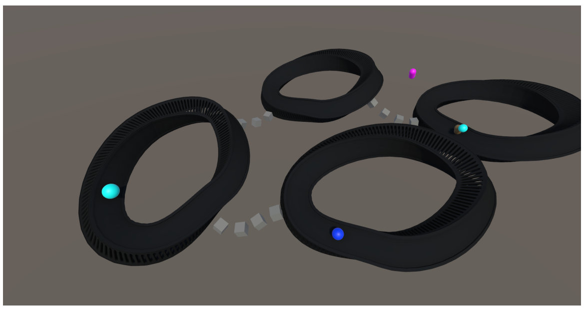
character collects new-shape item



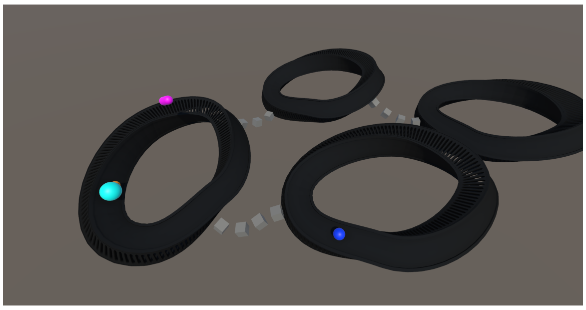
player places new shape



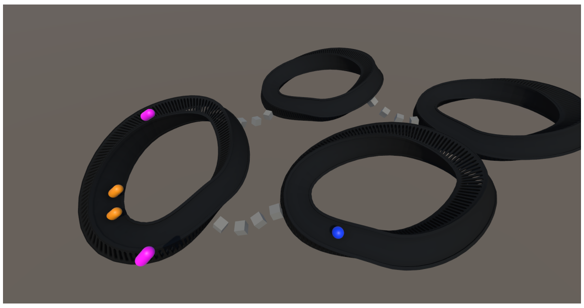
each shape checks whether it should spawn a new item, then 2 energy-items & 1 new-shape-item are spawned



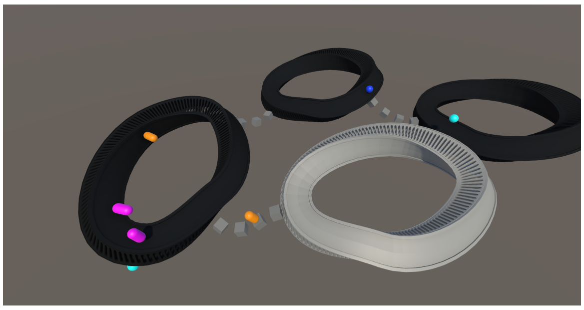
the player bridges and walks over to the energy-item



As the other energy item was around for some time, it became quite big (=much energy) before the player collects it



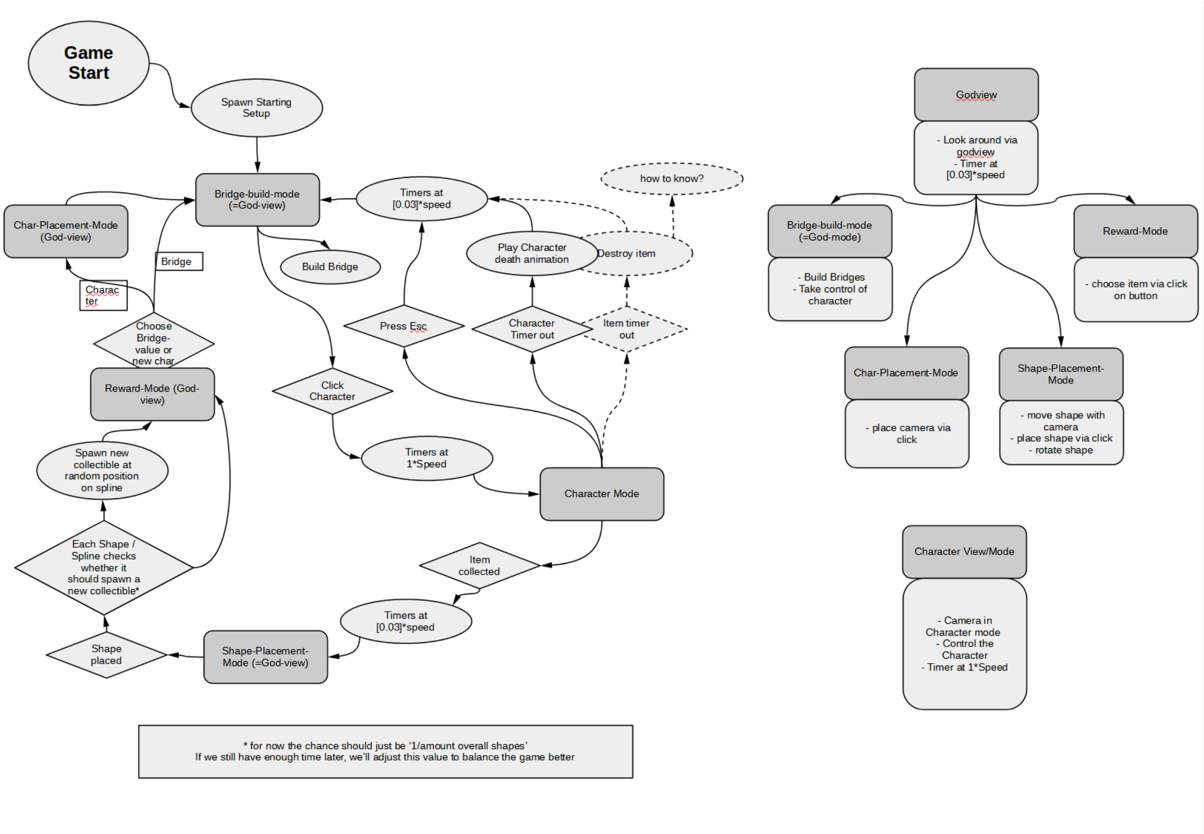
the over-energised character split into two character (see Energy-document)



the player was too slow at collecting the new-shape item -> the shape died and some new items spawned.

Mechanics

Game-States



Same Graphics, but bigger:
https://docs.google.com/document/d/1D-uIz0dpzgxvozispf0aC984RY6t_vw6-EdqBH8vJa8/edit?usp=drive_link

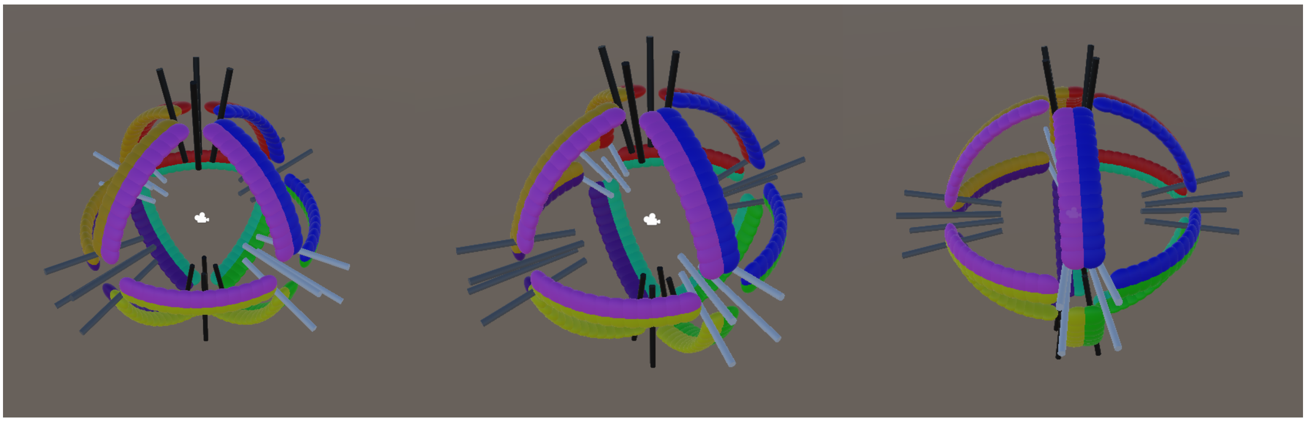
Camera

Explanation:

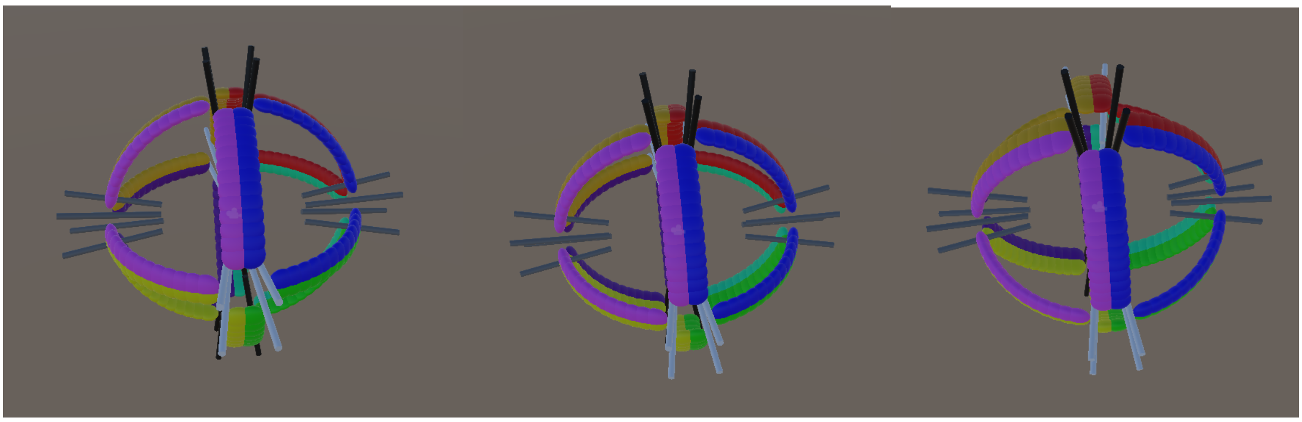
The player can see the whole environment; They can adjust the camera in multiple ways
A) Zoom
B) Pan
C) TIlt
D) Roll
D) Further Functions in “Godmode-Camera Advanced Functions”
(Ignore until the basic functions are done & until the “Godmode-Camera Advanced Functions” actually lands in the To-Do Folder)

Example:

Zooming via mouse-wheel (towards / away from the pivot object):



Rotating (Pan & Tilt) around an object:
Right-clicking (on the pivot object) and moving the mouse towards the left / right rotates the camera. The amount of rotation is based on the distance from the original right-clicking-point. (e.g. Right click on Screenposition (980, 540) then moving to (1180, 540) rotates the camera around the pivotobject to the position calculated by (startRotation+(1180-960)*([rotationSpeed]))



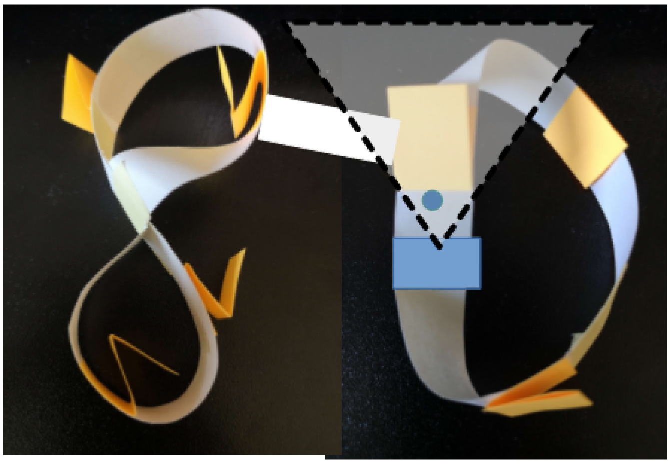
Same for upwards / downwards rotation

The rotation of the camera upwards / downwards should also work for more than 90° up/down, without the left/right rotation being mirrored when up/down is rotated more than 90°.

Rotation (Roll):
If the player right-clicks (&drags) at the edge of the screen, the camera does a roll-rotation

Character Camera shows only a part of the whole thing;
There are two options for this; I'd like to try both (=pls create option A first, afterwards option B with a public bool that let's us switch between the options during playtest):

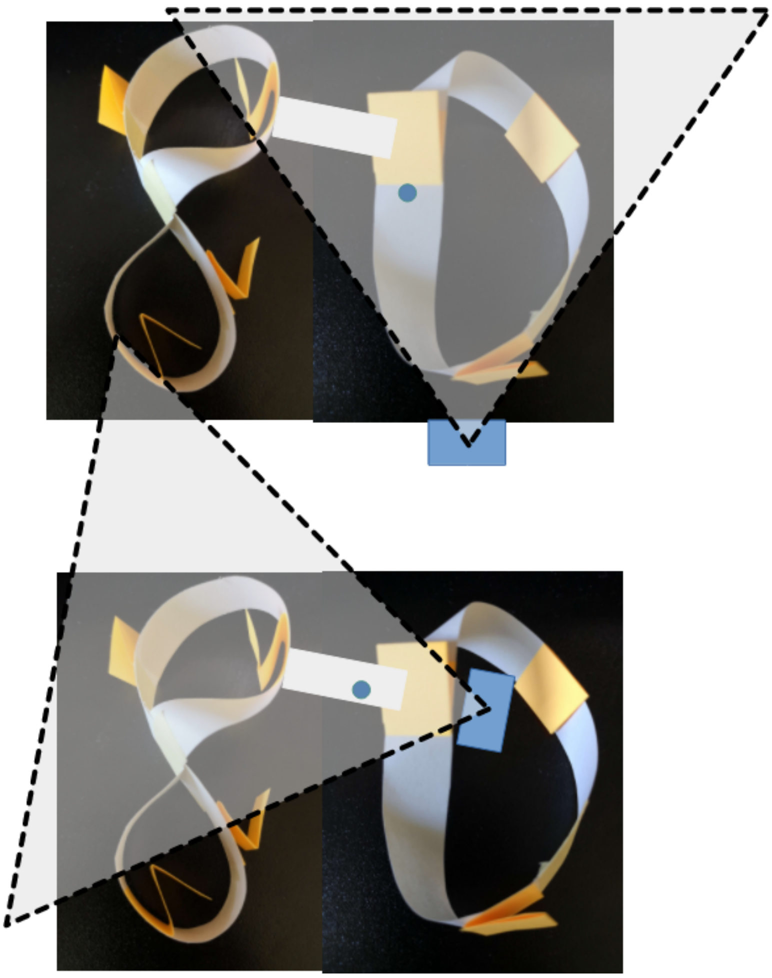
Option A, Character zoomed:



Camera closely behind the character (pls softcode the distance)

Character turns / moves = camera follows

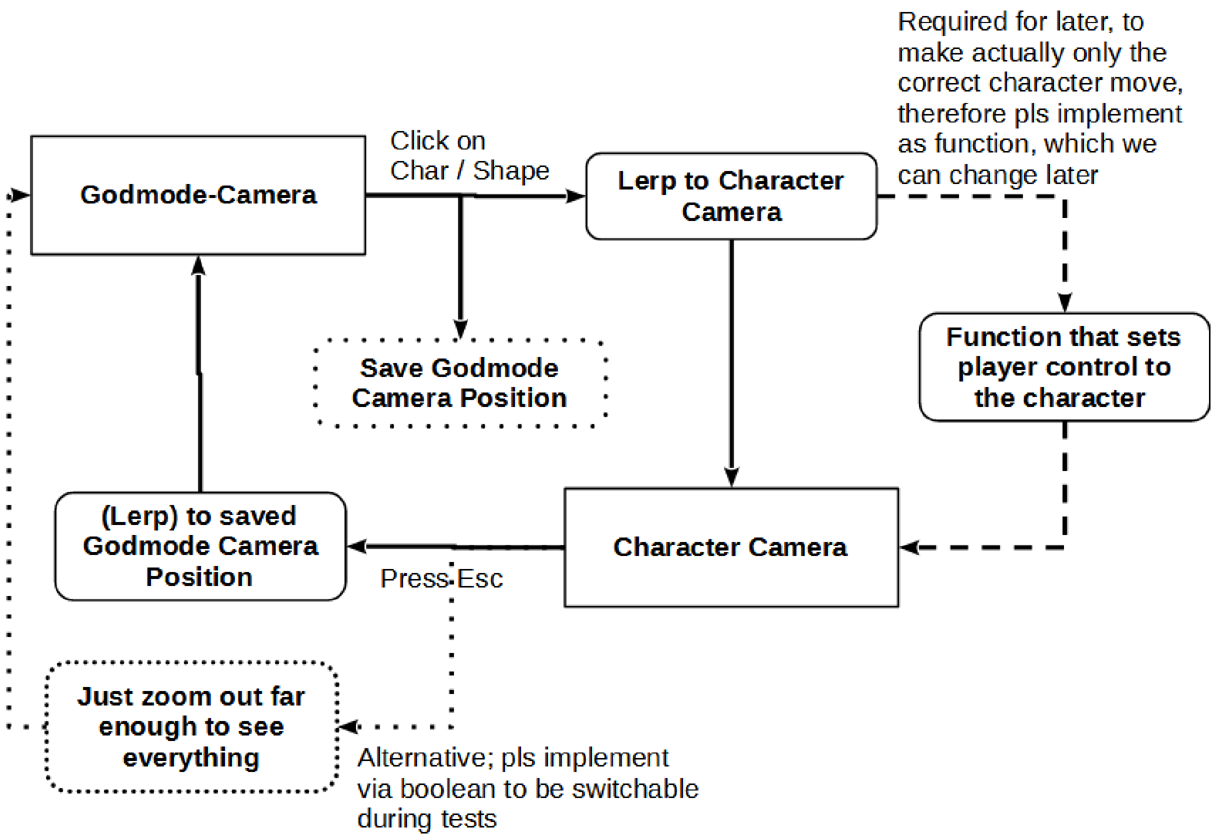
Option B, Shape zoomed *(not implemented into the game):*



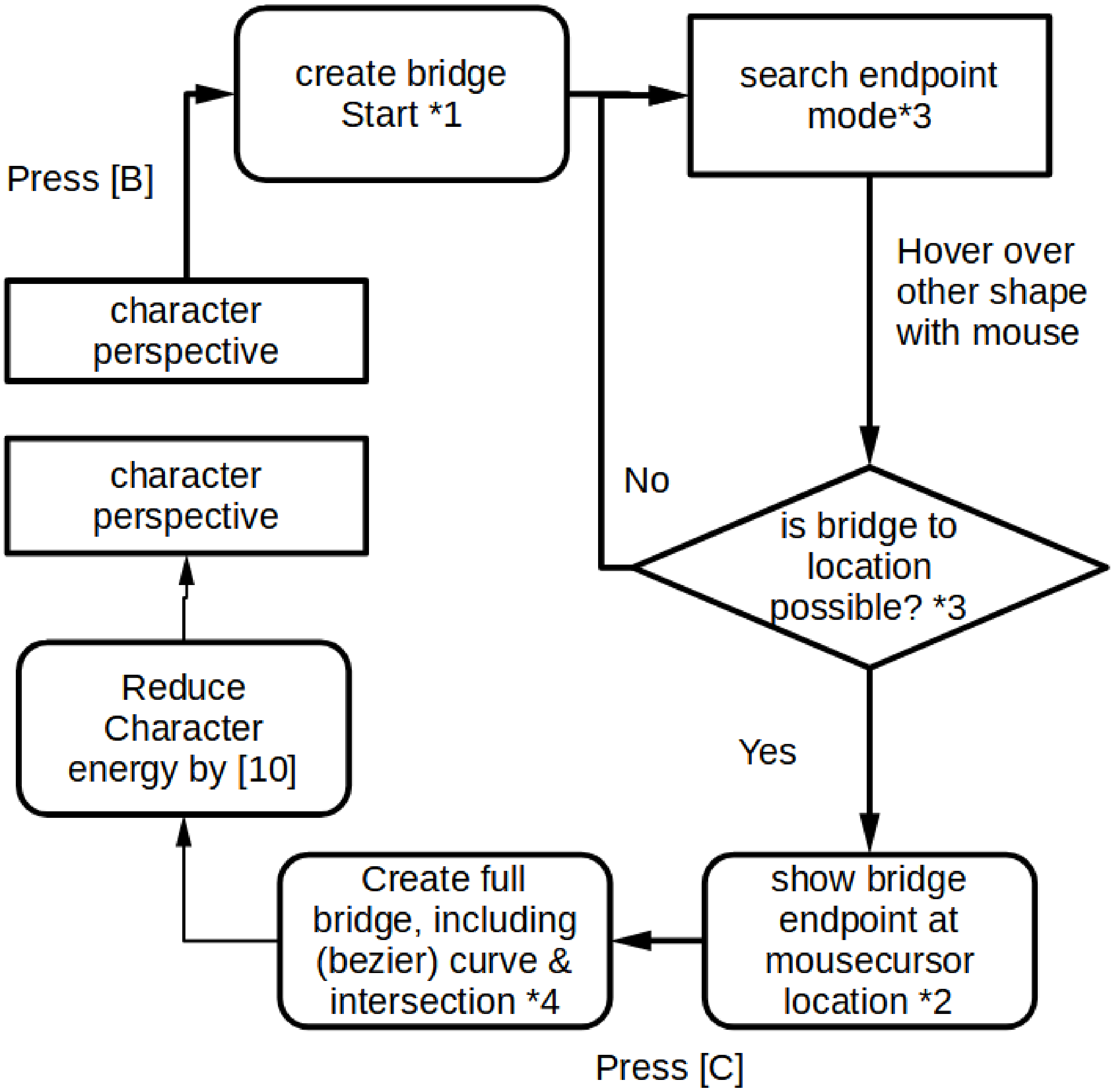
The camera zooms in, so that it sees the whole shape at once

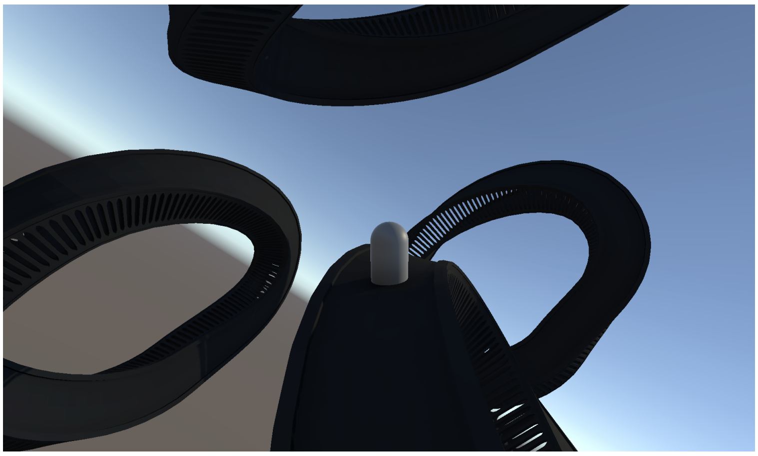
Character walks onto bridge / other shape -> Camera moves/rotates to new position to see the bridge/other shape

Switching Cameras

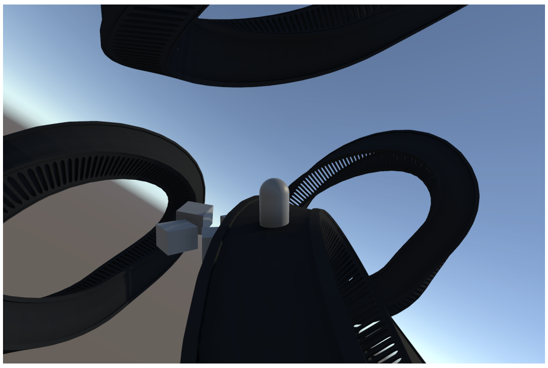


Building Bridges

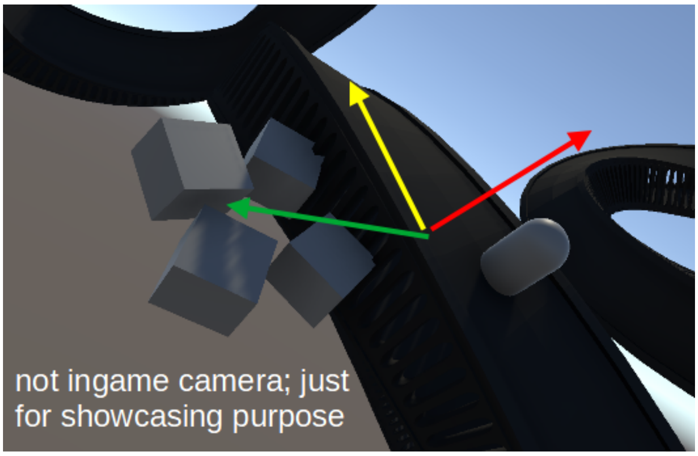




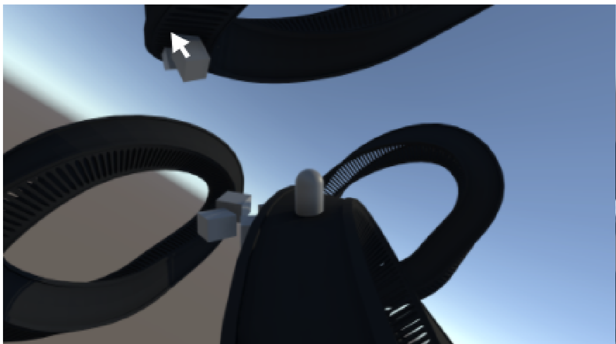
*0: The player takes control over a character
-> Character Camera (see camera document)

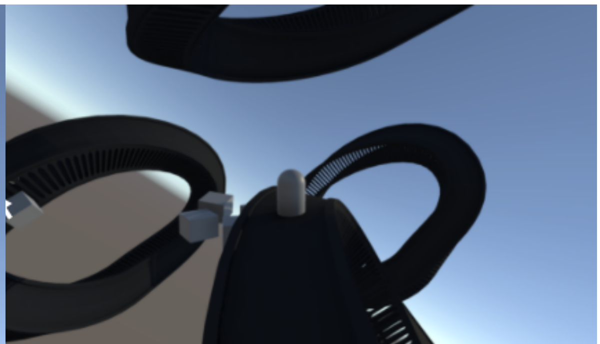


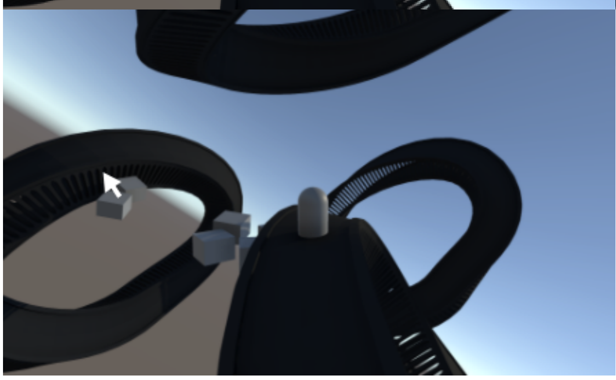
*1 The Player starts building a bridge;
The start-point & end-point of bridges are going sideways off the path



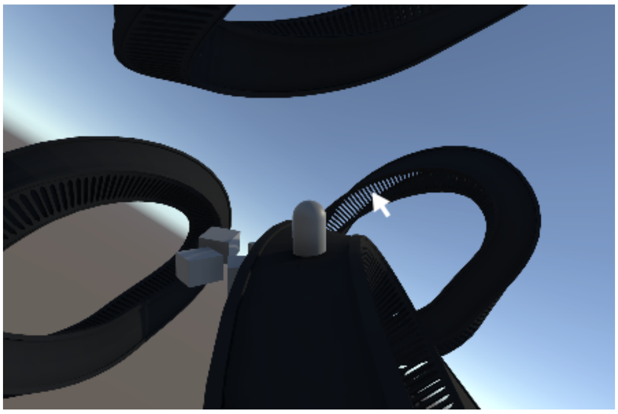
Red = normal of the curve
Yellow = direction the curve leads to
Green = orthagonal to the other two => starting vector of the bridge



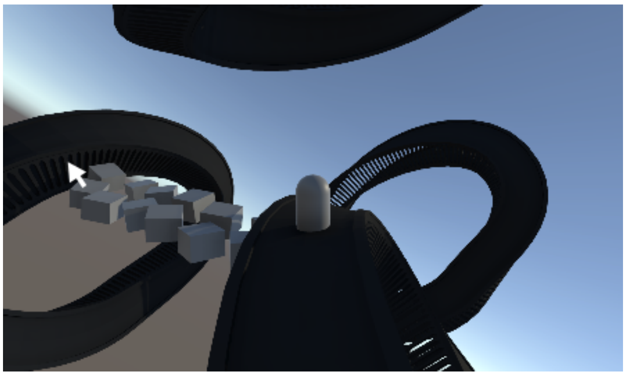




*2 Hovering with the mouse shows bridge end locations



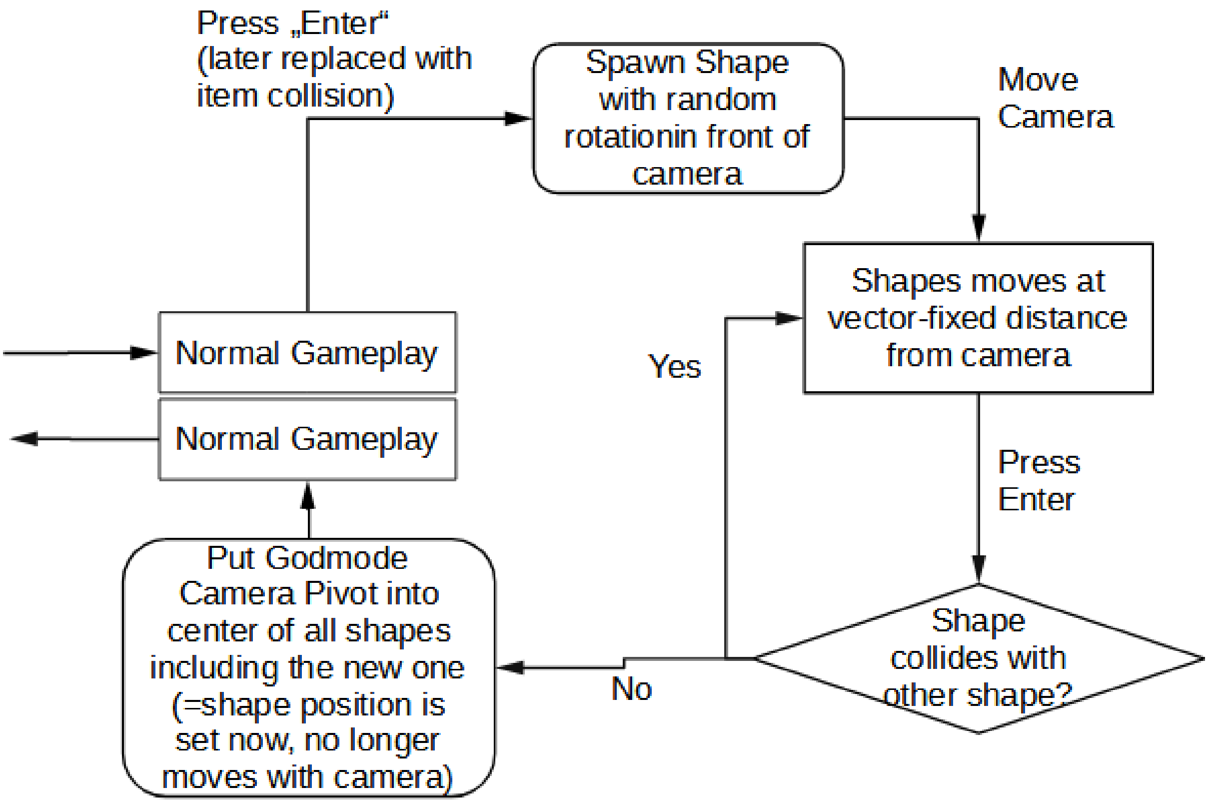
*3 Some location (e.g. too far away; or tilted in a too different direction [=whatever is too hard to implement]) can't be reached with a bridge, so no other end of the bridge is foreshadowed

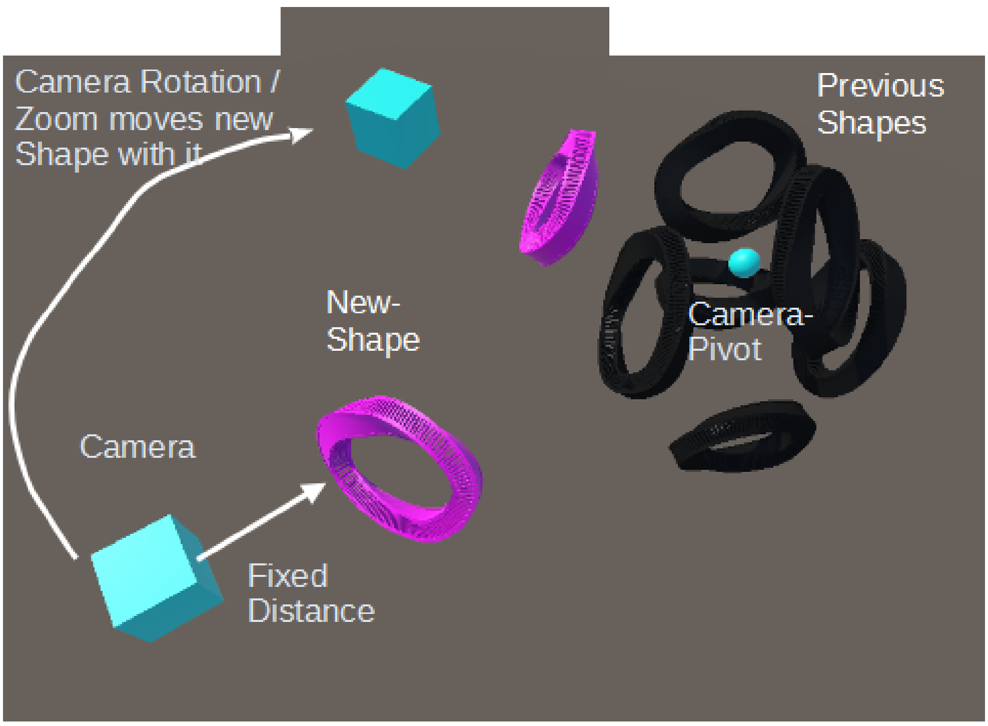


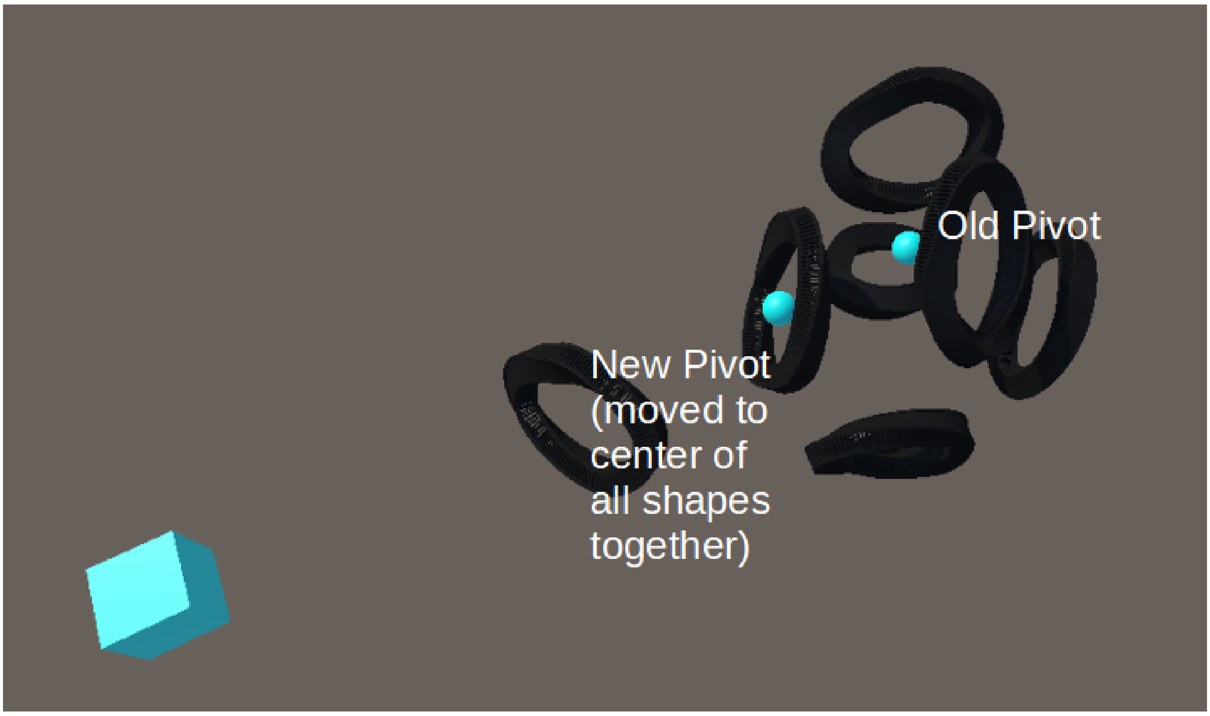
*4 Creating a bridge = creating a curve & some objects along (& directly next to) the curve, that are facing according to the normals of the bridge at that point

Portals first -> then bridges

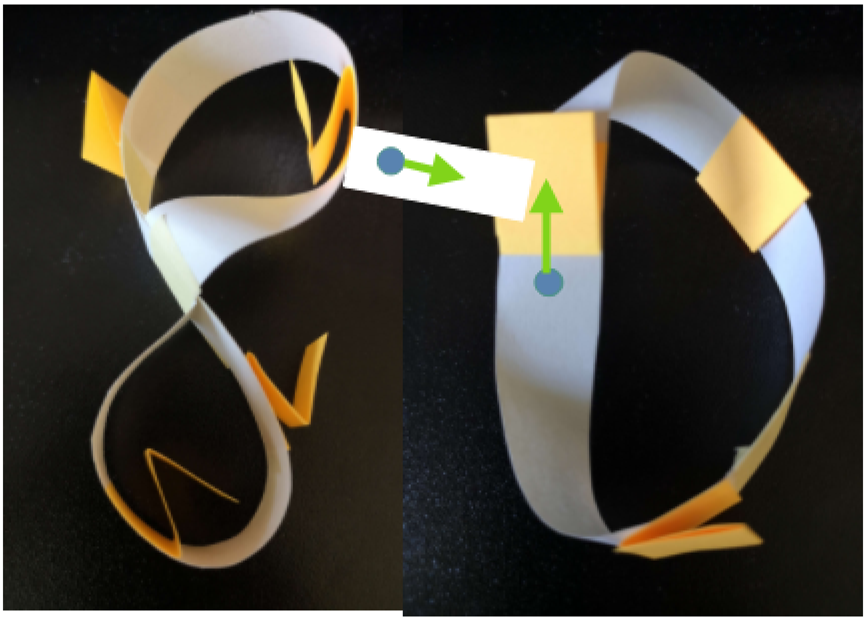
Shape Placement





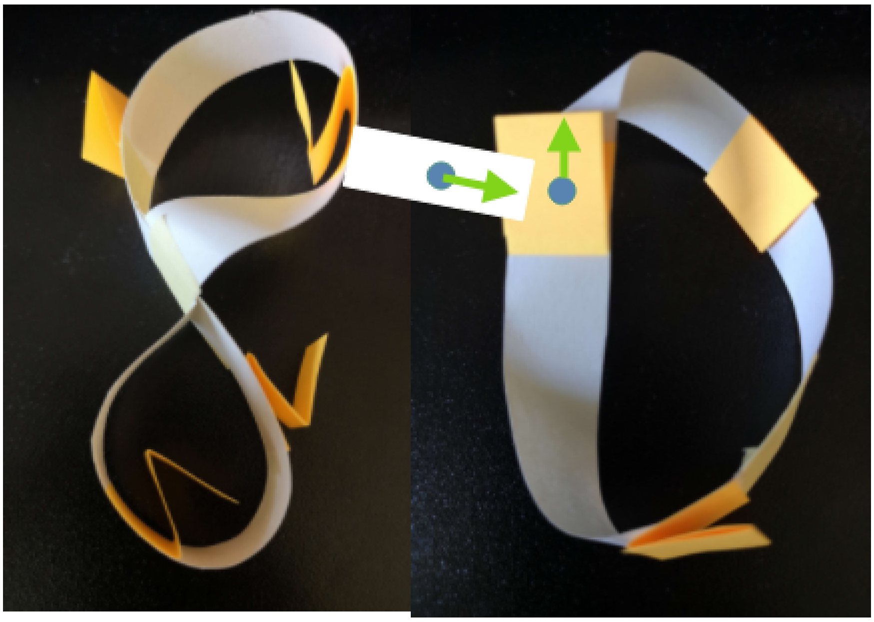

Old Pivot
New Pivot
(moved to
center of
all shapes
together)

Character Movement

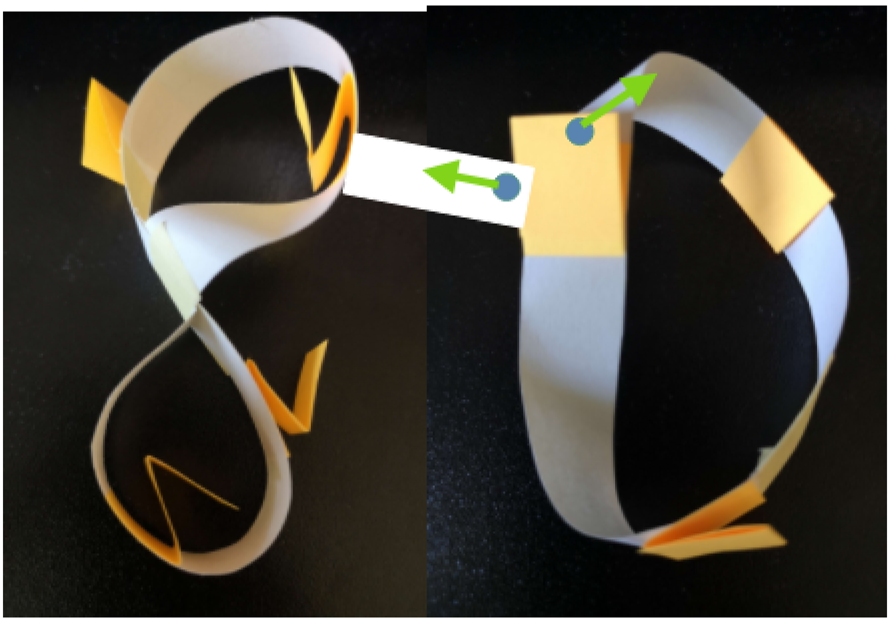


Walk Along Path:

The Characters (blue) move automatically along their path (=along the bezier curve, which they are on);

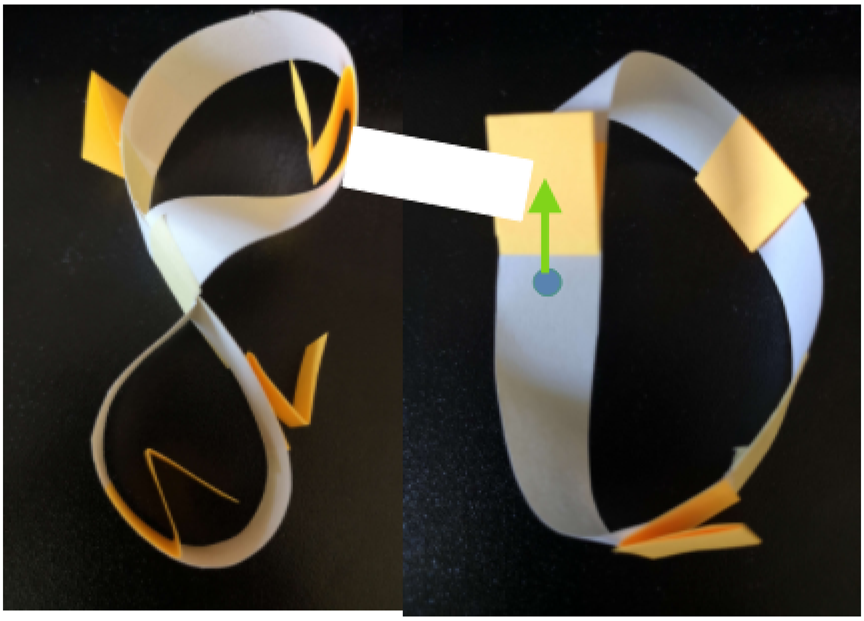


Figures don’t take any turns at intersections, they always go straight ahead.

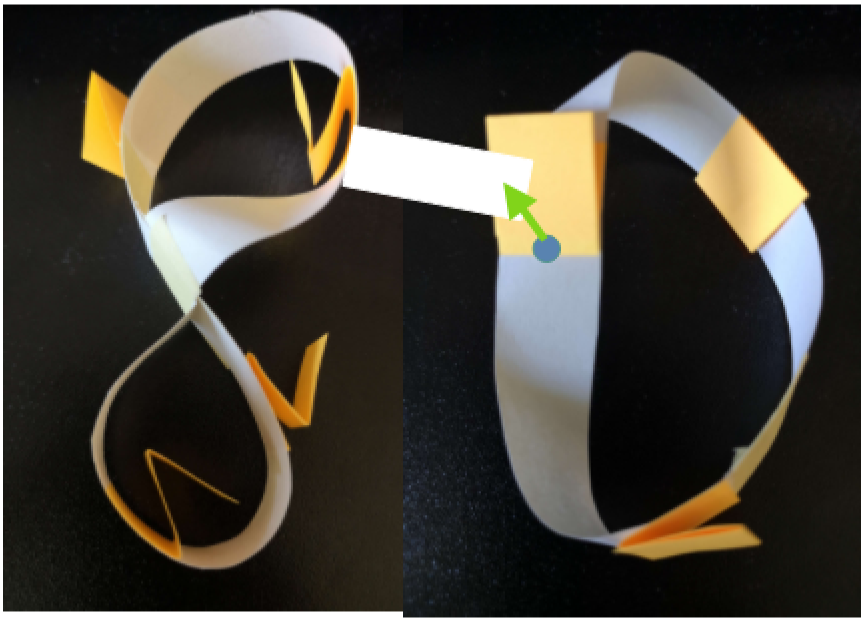


When they are on a not-circular Bézier path and reach the end, they turn around and walk towards the other direction.

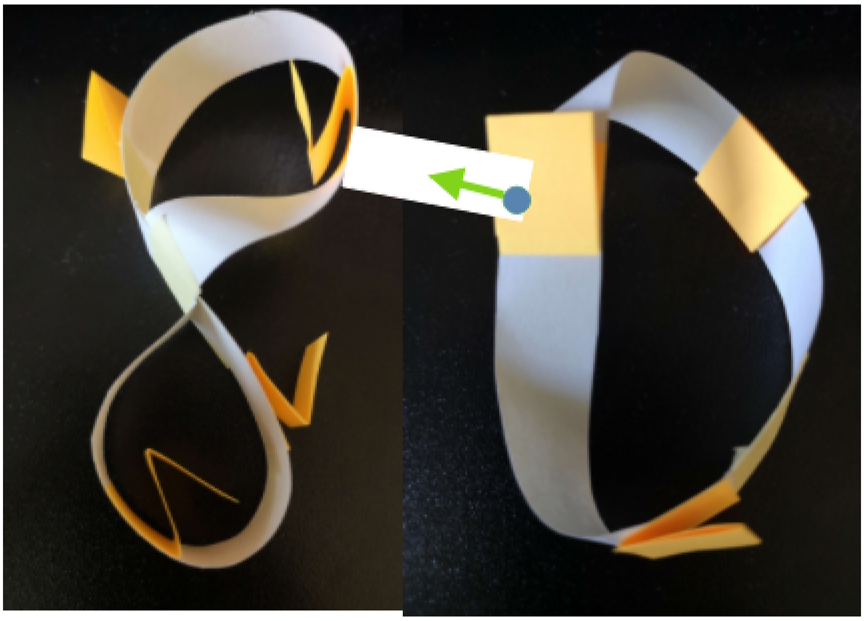
Switch Direction



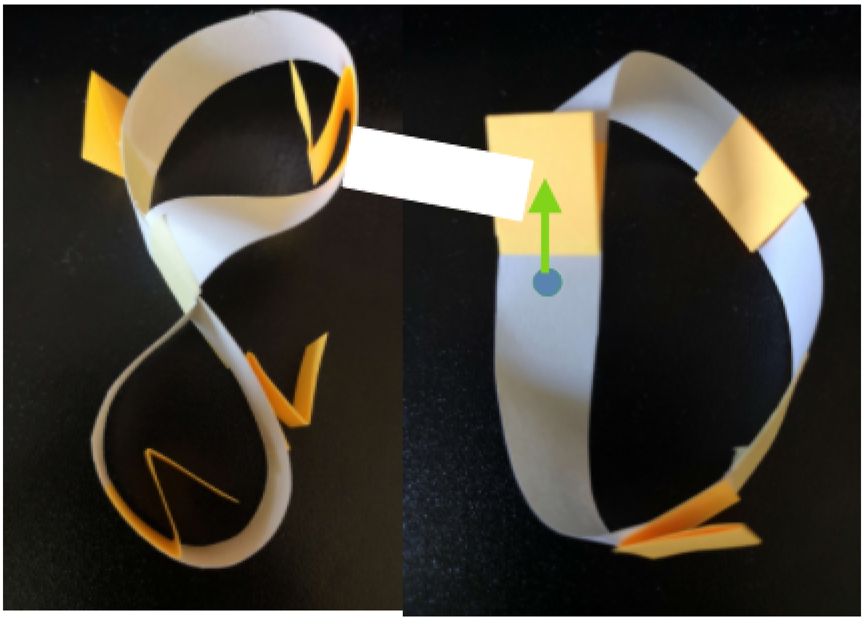
Upon taking control over a character, the player can rotate them.



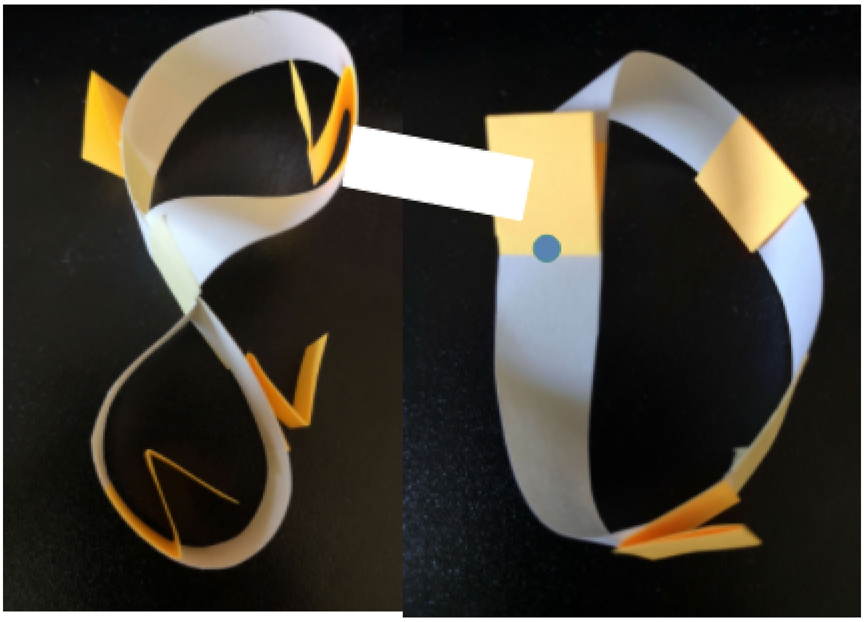
Pressing the rotation button A/D while being on an intersection area (yellow on the image) lerps the characters position from where they are to the point closest on the other bezier curves.

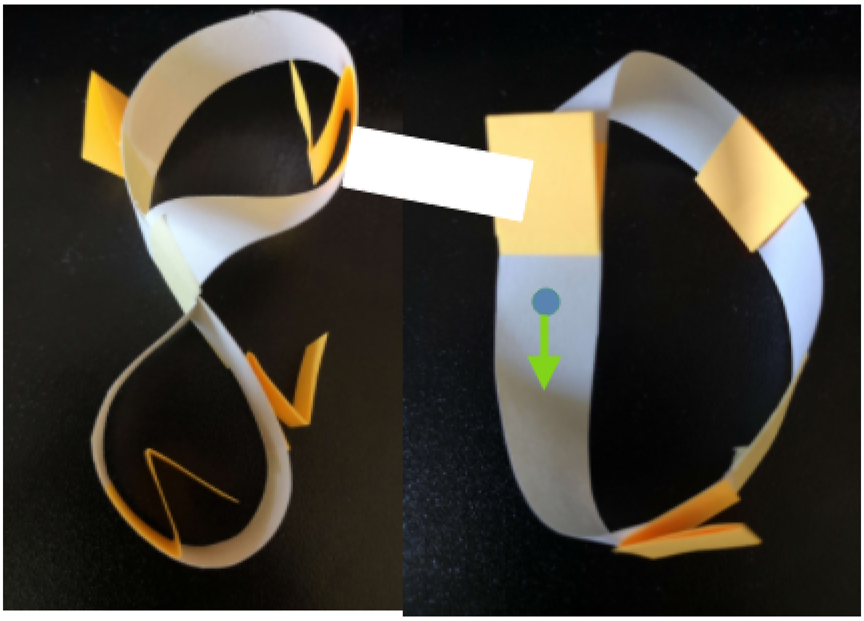


Then the character walks along the new bezier curve.

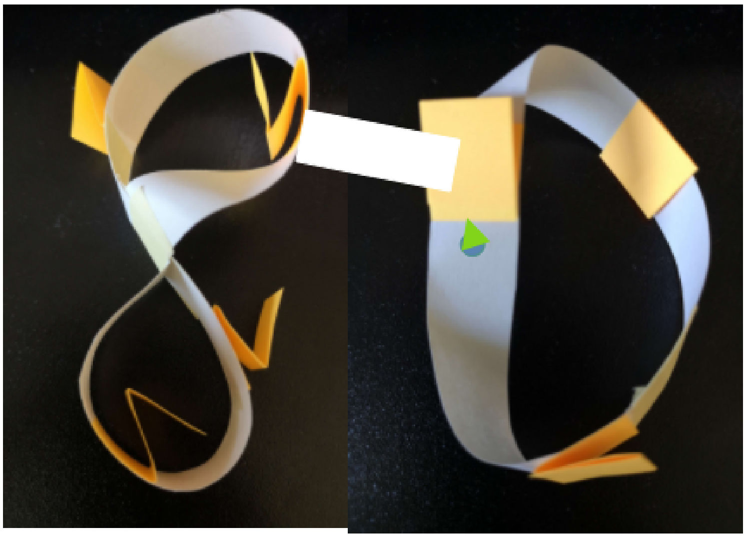


Pressing S while controlling a character turns the character around by 180°, so that they walk along the bezier curve in the opposite direction. [0.5sec]



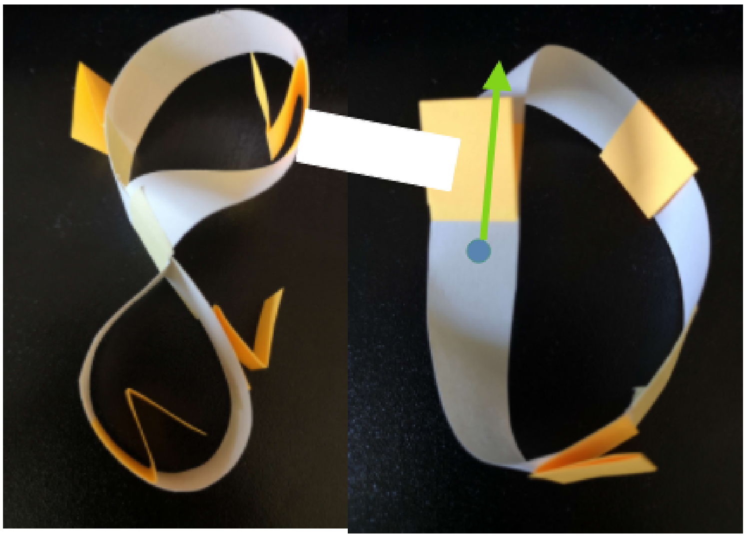


Merge-Speed mode

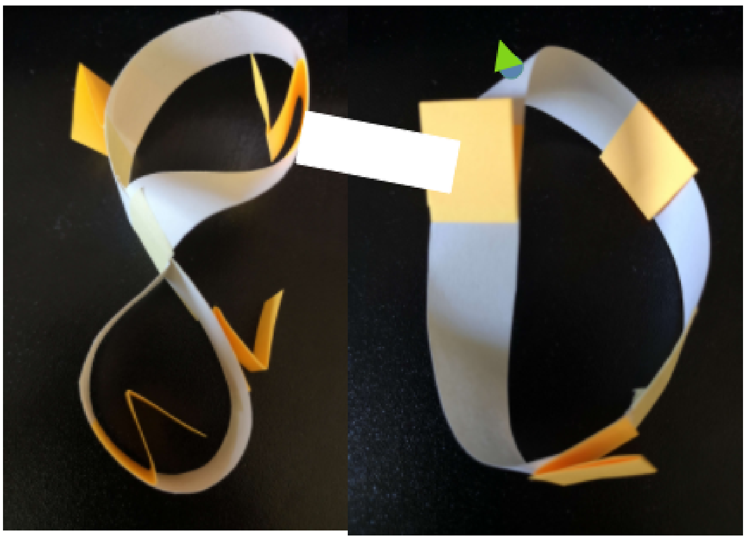


(not implemented into the game, due to time-constraints & because changing the speed between game-states also solves the speed problem)

Pressing W merges the character with the ground, which takes [~0.3sec (=animation starting time)].
Player stops in that time.



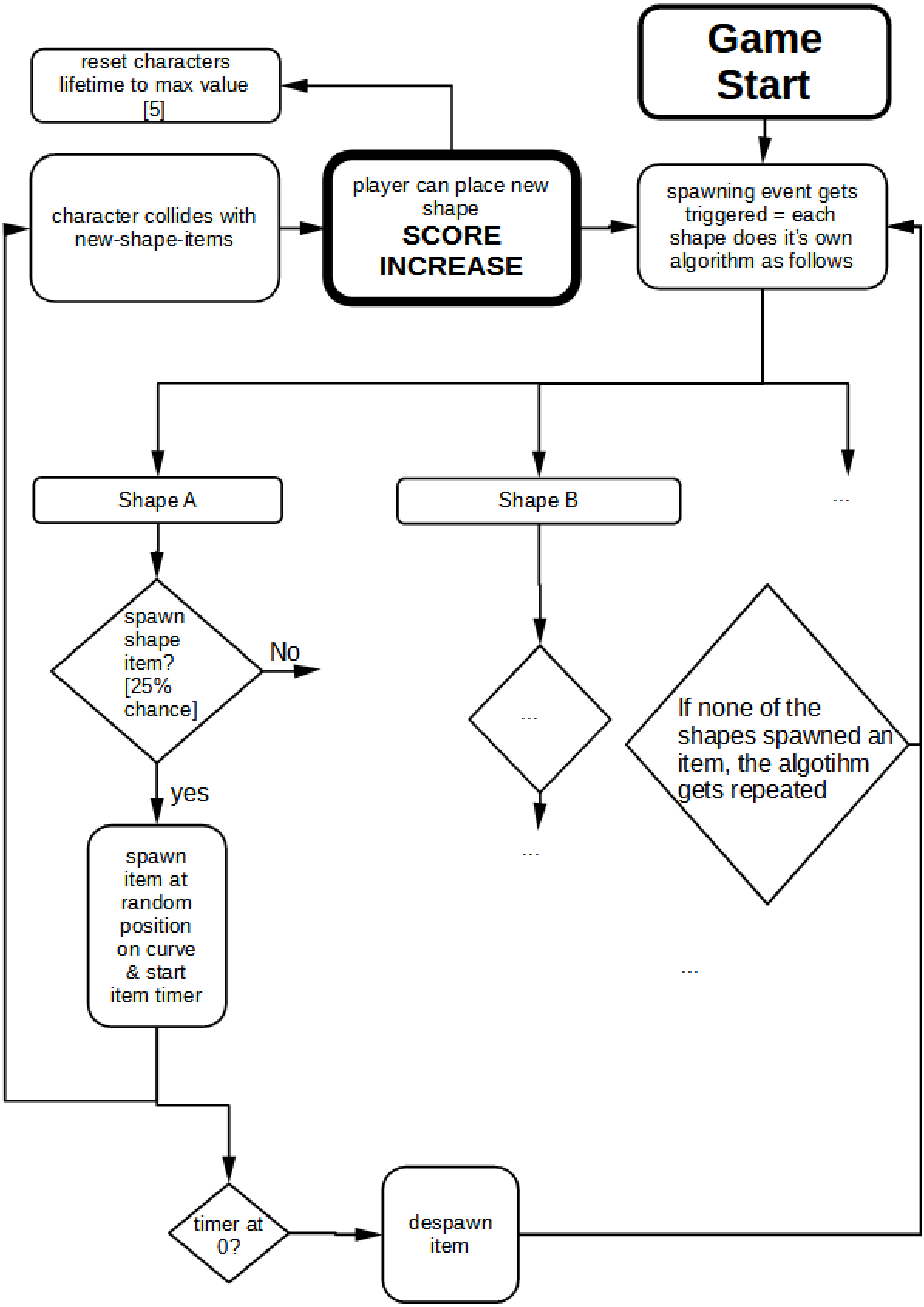
Afterwards the character moves at highspeed ([2*speed]) through the ground.



Pressing W again unborrows the character, so that it walk normally again.
[~0.3s]
Player stops in that time.

While any of the movement types are active, the player can’t do another movement type (e.g. while the character rotates, it can’t speed up; While it is in merge-speed mode, it can’t rotate (only after unmerging it can rotate again).

Items



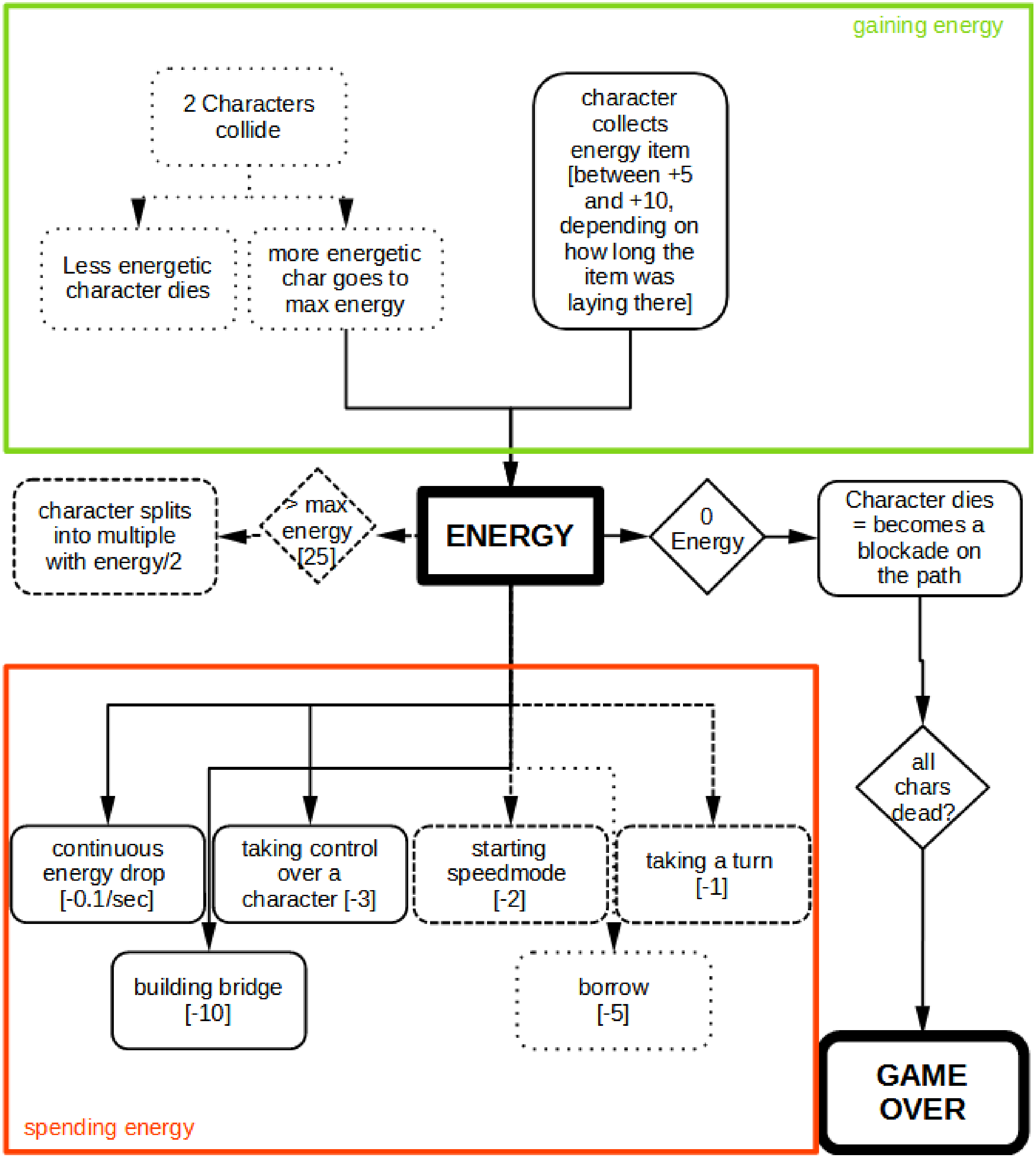
Game Start
spawning event gets triggered = each shape does it's own algorithm as follows
player can place new shape
SCORE INCREASE
reset characters lifetime to max value [5]
character collides with new-shape-items
Shape A
Shape B
...
spawn shape item? [25% chance]
No
yes
...
If none of the shapes spawned an item, the algotihm gets repeated
...
...
spawn item at random position on curve & start item timer
timer at 0?
despawn item

Energy

This mechanic didn't get implemented due to missing time & also because it potentially adds too much unnecessary complexity to the game

Everything the player does in the game is based on energy

Each one of the characters has their energy, which is refilled / depleted by the following (inter)actions:



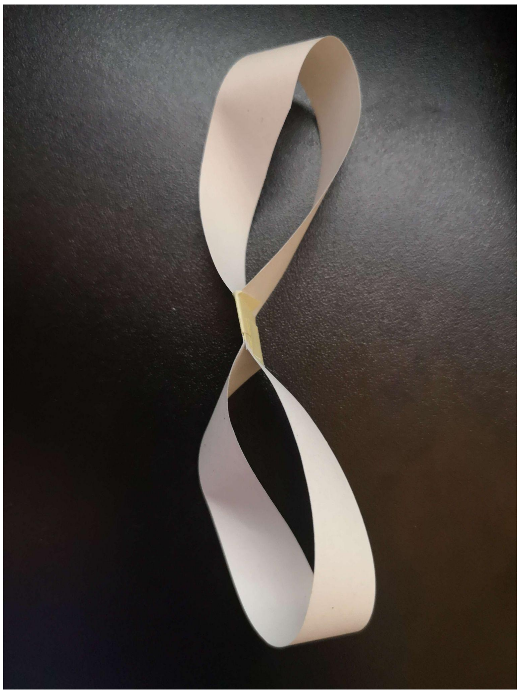
Art

Original Paper Strips to Model:

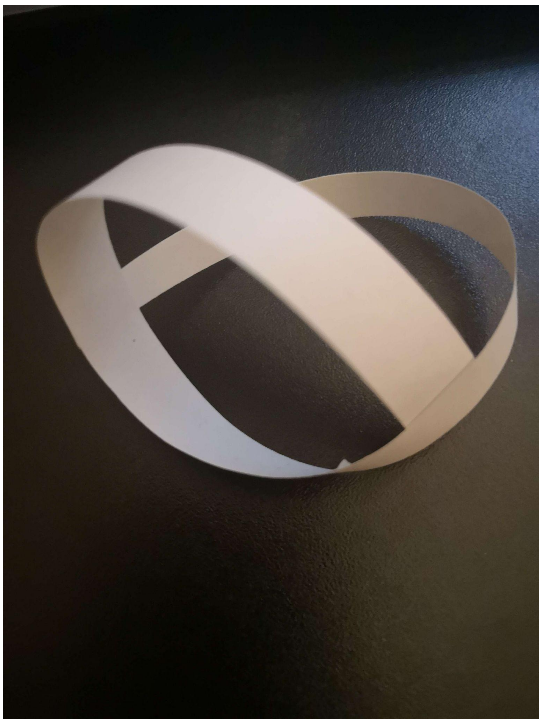
Möbius

Figure-8 *(later replaced by “Chair”)*

Through the Loop

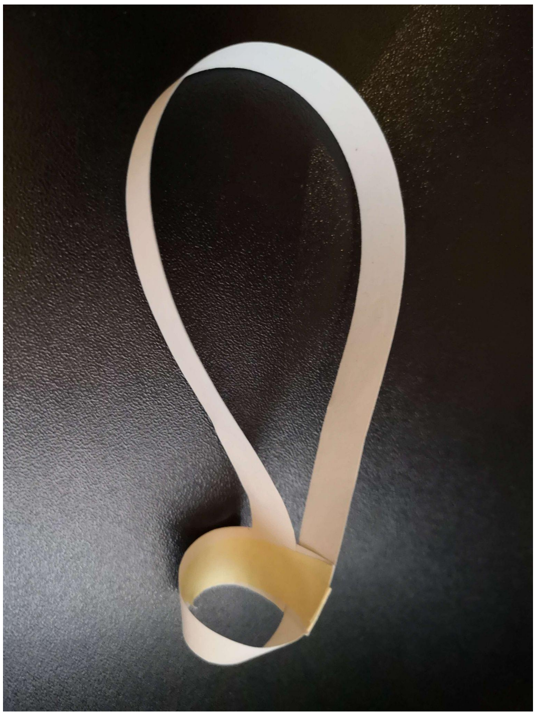


Butterfly

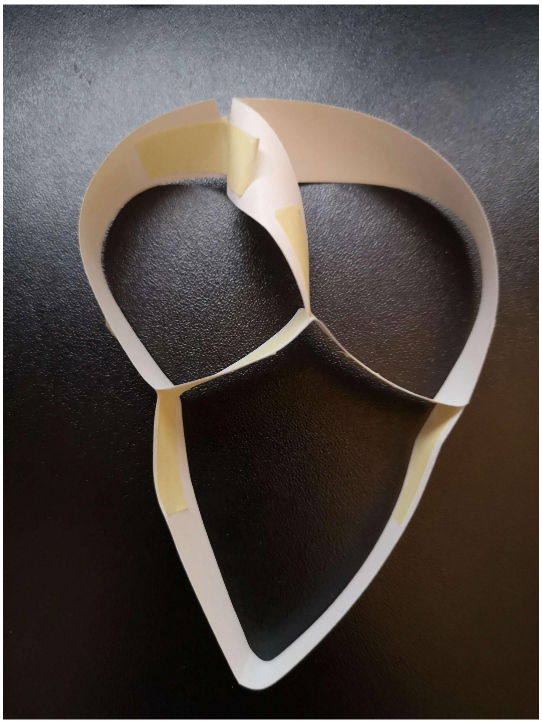


Roundabout

Pretzel



Skyscraper



Twisted Heart

Note to Skyscraper: not in the game due to bugs when spawning items close to it
Note to Twisted Heart: Got Replaced by the Frog-Spline Later

Movement Animations -> see “Character Movement”